

HDOC - Day 13 Activity

C Programming Assignment

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Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2
- Find the sum of its even digits and the sum of its odd digits separately.

E.g. Number: 12345, Sum_even = 6, Sum_odd = 9 **Algorithm**

1. START
2. Input a positive integer n.
3. If $n \leq 0$, display an error message and STOP.
4. Display the menu:
1 - Count even and odd digits 2 -
Find sum of even and odd digits 5.
Input choice.
6. Set $temp = n$.
7. Use `switch(choice)`.
8. If $choice = 1$: $even_count = 0$, $odd_count = 0$ WHILE $temp > 0$: $digit = temp \% 10$
IF $digit \% 2 == 0$:
 $even_count = even_count + 1$
ELSE: $odd_count = odd_count + 1$
 $temp = temp / 10$
Display $even_count$ and odd_count .

9. If choice = 2: sum_even = 0, sum_odd = 0
 WHILE temp > 0: digit = temp % 10 IF digit
 % 2 == 0:
 sum_even = sum_even + digit
 ELSE: sum_odd = sum_odd +
 digit temp = temp / 10
 Display sum_even and sum_odd.
10. Otherwise, display "Invalid choice".
11. STOP

Test Case Table

Test	Input(n)	Choice	Expected Output	Result
1	12045	1	Even digits= 3 , Odd digits= 2	Pass
2	12345	2	Sum even= 6 , Sum odd= 9	Pass
3	24680	1	Even digits= 5 , Odd digits= 0	Pass
4	13579	2	Sum even= 0 , Sum odd= 25	Pass
5	12345	3	Invalid choice	Pass
6	-25	1	Enter a positive integer only.	Pass

C Code

```
#include <stdio.h>
int main() {
    int n, temp, digit, choice; int
    even_count = 0, odd_count = 0; int
    sum_even = 0, sum_odd = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Enter a positive integer only.\n"); return
        0;
    }
    printf("\nMENU\n"); printf("1. Count even and
    odd digits\n"); printf("2. Find sum of even and
    odd digits\n"); printf("Enter your choice: ");
    scanf("%d", &choice); temp = n; switch (choice)
    { case 1:
```

```
while (temp > 0) {
    digit = temp % 10;
    if (digit % 2 == 0)
        even_count++; else
        odd_count++;
    temp = temp / 10;
}
printf("Even digits = %d\n", even_count);
printf("Odd digits = %d\n", odd_count);
break; case 2:
while (temp > 0) {
    digit = temp % 10;
    if (digit % 2 == 0)
        sum_even += digit;
    else sum_odd +=
    digit;
    temp = temp / 10;
}
printf("Sum even = %d\n", sum_even);
printf("Sum odd = %d\n", sum_odd); break;
default: printf("Invalid choice\n");
}
return 0;
}
```

Compilation , Execution and Testing Cases

```
Session Actions Edit View Help
(kali@kali)-[~]
$ vi evenodd.c
(kali@kali)-[~]
$ cc evenodd.c
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12045

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 3
Odd digits = 2

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum even = 6
Sum odd = 9

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 24680

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 5
Odd digits = 0
```

```
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 0
Odd digits = 5

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum even = 0
Sum odd = 25

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 3
Invalid choice

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: -25
Enter a positive integer only.
```